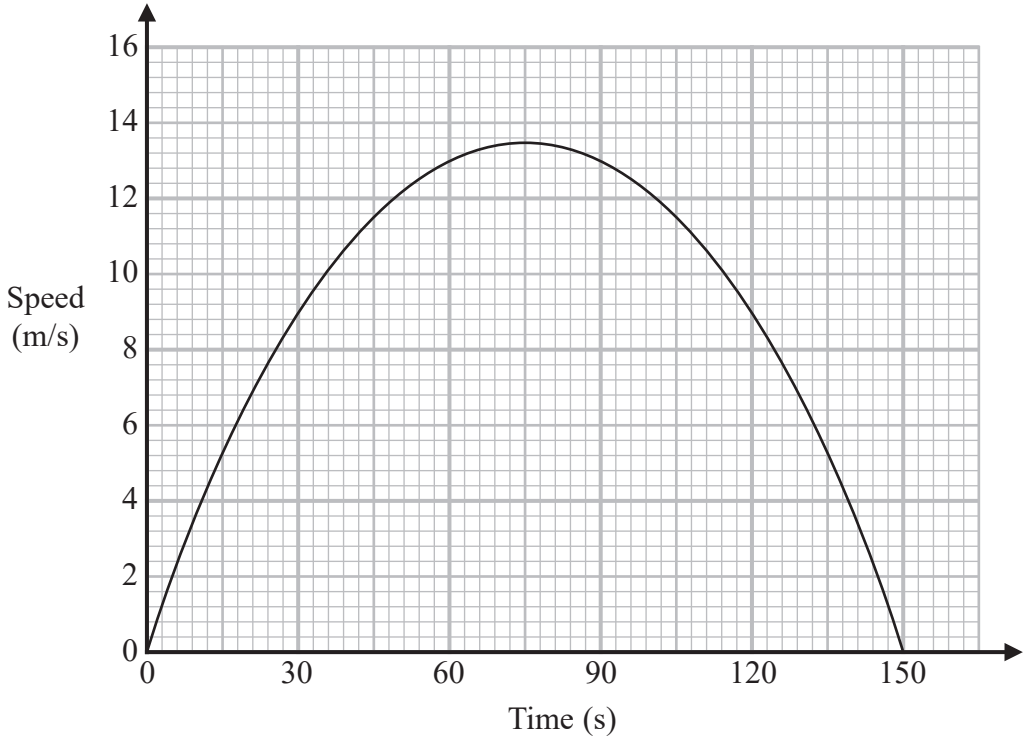
16 Here is a speed-time graph for a car.



(a) Work out an estimate for the distance the car travelled in the first 30 seconds.

At 30 s the speed is about 9 m/s.

Triangle estimate = ½ × 30 × 9 = 135 m.

2 marks total
Area under speed graph
135 m

..... m

(2)

(b) Is your answer to part (a) an underestimate or an overestimate of the actual distance the car travelled in the first 30 seconds?
Give a reason for your answer.

Underestimate: the curve lies above the straight

triangle edge, so extra area is omitted.

1 mark
Curve above triangle
Underestimate

(1)

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Julian used the graph to answer this question.

Work out an estimate for the acceleration of the car at time 60 seconds.

Here is Julian's working.

$$\text{acceleration} = \text{speed} \div \text{time}$$

$$= 13 \div 60$$

$$= 0.21\dot{6} \text{ m/s}^2$$

Julian's method does not give a good estimate of the acceleration at time 60 seconds.

(c) Explain why.

13/60 is average acceleration from 0 to 60 s.

Instantaneous acceleration at 60 s is the tangent gradient.

1 mark
Use tangent at 60 s
Not speed divided by time

(1)

(Total for Question 16 is 4 marks)

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